

## Chapter 1: Basic Network Concepts

### 1. Explain client-server model & peer-to-peer model with a suitable diagram. Write advantages & disadvantages.

Client-Server Model:

- A centralized network where a server provides resources/services to multiple clients.
- Advantages: Centralized control, better security, efficient data management.
- Disadvantages: High cost, requires maintenance, server failure affects all clients.

Peer-to-Peer (P2P) Model:

- A decentralized network where each device acts as both client and server.
- Advantages: Cost-effective, easy setup, no need for a dedicated server.
- Disadvantages: Less secure, difficult to manage, slower performance for large networks.

(Diagram required for both models.)

### 2. Compare OSI & TCP/IP model, stating advantages & disadvantages.

#### OSI Model | TCP/IP Model

|                 |                                                                                     |
|-----------------|-------------------------------------------------------------------------------------|
| Layers          | 7 Layers   4 Layers                                                                 |
| Developed by    | ISO   DARPA                                                                         |
| Usage           | Theoretical model for networking   Practical model used in real-world communication |
| Advantages      | Clear distinction of functions   Simpler, widely implemented                        |
| ,highly modular |                                                                                     |
| Disadvantages   | Complex, rarely used in   Not as modular as OSI                                     |
| practice        |                                                                                     |

(Mention 4 points as required.)

### 3. Explain physical, logical, port, and socket address.

- Physical Address: Also called MAC address, uniquely identifies a device in a network.
- Logical Address: An IP address assigned to devices for communication over a network.
- Port Address: A unique number assigned to a process for communication (e.g., port 80 for HTTP).
- Socket Address: Combination of IP address and port number, used to establish a connection.

#### 4. Explain various types of networks with suitable diagrams.

- LAN (Local Area Network): Small network within a single location (e.g., office, home).
- MAN (Metropolitan Area Network): Covers a city or large campus (e.g., cable TV network).
- WAN (Wide Area Network): Covers large distances (e.g., the internet).
- PAN (Personal Area Network): Small network for personal devices (e.g., Bluetooth).

(Diagram required for each.)

#### 5. List any 4 applications of computer networks.

1. File sharing
2. Remote access
3. Internet services (email, web browsing)
4. Online communication (VoIP, video conferencing)

#### 6. Explain the working of OSI model with a suitable diagram. Describe roles and responsibilities of each layer.

The OSI model consists of 7 layers:

1. Physical Layer: Transmits raw data bits over physical medium.
2. Data Link Layer: Ensures error-free data transfer between nodes.
3. Network Layer: Handles IP addressing and routing.
4. Transport Layer: Ensures reliable data delivery (TCP, UDP).
5. Session Layer: Manages and maintains connections.
6. Presentation Layer: Translates data formats, encryption, compression.
7. Application Layer: Interfaces with user applications (HTTP, FTP, SMTP).

## Chapter 2: Network Topologies and Networking Devices

### 1. Calculate the number of cables required for different topologies.

- Bus Topology: Requires  $(n-1)$  cables for 'n' devices.
- Star Topology: Requires 'n' cables (each device connects to a central hub/switch).
- Ring Topology: Requires 'n' cables (each device connects to two neighbors).
- Mesh Topology: Requires  $(n(n-1))/2$  cables for full connectivity.

### 2. Differentiate between Hub & Switch.

#### Hub | Switch

- | Function | Broadcasts data to all devices | Sends data only to the intended device |
- | Speed | Slower due to network congestion | Faster due to efficient switching |
- | Security | Less secure (anyone can access data) | More secure (data is directed) |
- | Efficiency | Inefficient in large networks | Highly efficient |

### 3. List and explain network topologies with suitable diagrams, characteristics, advantages & disadvantages.

- Bus Topology: Single backbone cable, easy to implement, but failure disrupts the network.
- Star Topology: Central hub connects all devices, reliable, but hub failure affects all.
- Ring Topology: Data moves in a circular path, efficient but one failure affects all.
- Mesh Topology: Every device connects to every other, very reliable but expensive.

### 1. Types of Network Topologies:

- Star Topology: All devices are connected to a central hub or switch. If the hub fails, the entire network is affected.
- Bus Topology: A single central cable (bus) connects all computers. If the main cable fails, the network stops working.
- Ring Topology: Devices are connected in a circular manner. Data travels in one or both directions.
- Mesh Topology: Every device is connected to every other device, providing redundancy and reliability.
- Tree Topology: A hierarchical structure with multiple levels of branches.
- Hybrid Topology: A combination of two or more different topologies.

## 2. Characteristics of Data Rate for Noiseless Channel (Shannon's Capacity Formula):

Shannon's theorem states that the maximum data rate for a noiseless channel is given by:

$$C = B [\log_2 (1 + S/N)]$$

Where:

- $C$  is the channel capacity (bits per second)
- $B$  is the bandwidth of the channel (Hz)
- $S/N$  is the signal-to-noise ratio

## 3. Factors Affecting Network Performance:

- Bandwidth: Higher bandwidth means better performance.
- Latency: The delay in data transmission affects performance.
- Packet Loss: Lost data packets can slow down communication.
- Network Congestion: Too much traffic can slow down the network.
- Hardware Limitations: Poor-quality routers and switches can affect performance.

## 4. Design a Network for an Application (Example):

- Example: College network with servers, routers, access points, and connected computers in different departments.

## 5. Data Transmission Impairments:

- Attenuation: Signal strength decreases over distance.
- Noise: External interference distorts signals.
- Latency: Delay in data transmission.
- Jitter: Variation in packet arrival times.

## 6. Network Devices and Their Functions:

- Hub: Connects multiple devices in a network. Can be active (powered) or passive (unpowered).
- Switch: A smarter hub that forwards data to specific devices.
- Bridge: Connects two networks and filters traffic.
- Router: Directs data between different networks.
- Gateway: Acts as a bridge between different network protocols.
- Repeater: Amplifies weak signals to extend the network range.

## Chapter 3

### 1. What is Switching?

Switching is how data moves from one place to another in a network. There are three types:

**Circuit Switching:** A direct path is created before communication starts (like phone calls).

**Packet Switching:** Data is broken into small parts (packets) and sent separately (like emails and web browsing).

**Message Switching:** Entire messages are sent from one node to another until they reach the destination (like postal mail).

### 2. What is Multiplexing?

Multiplexing means combining multiple signals to send them over a single communication channel.

Types:

**FDM (Frequency Division Multiplexing):** Each signal is sent on a different frequency (like FM radio).

**WDM (Wavelength Division Multiplexing):** Used in fiber optics, where different light wavelengths carry different signals.

**TDM (Time Division Multiplexing):** Each signal gets a specific time slot (like sharing a road with different time schedules).

### 3. Bluetooth Technology

Bluetooth allows wireless communication between devices like phones, headphones, and smartwatches.

How It Works:

A piconet is formed, where one device (master) connects with others (slaves).

Multiple piconets together create a scatternet.

Uses short-range radio waves for communication.

### 4. What is Wi-Fi?

Wi-Fi allows devices to connect to the internet without wires.

Uses radio waves to send and receive data.

Works on 2.4 GHz and 5 GHz frequencies.

Common in homes, offices, and public places.

## 5. What is Guided Media?

Guided media means using physical cables to send data.

Types:

1. Twisted Pair Cable – Two copper wires twisted together (used in LAN networks).
2. Coaxial Cable – A single copper wire with insulation (used for cable TV).
3. Fiber Optic Cable – Uses light signals (used for fast internet).

## 6. Why Do We Need Transmission Media?

Transmission media is important because:

It carries data from one place to another.

It connects computers, phones, and other devices.

It affects speed, security, and quality of communication.

## 7. Factors Affecting Transmission Media

1. Speed – Faster cables mean better performance.
2. Interference – Wireless signals can be disturbed by other devices.
3. Cost – Fiber optic cables are expensive but fast.
4. Security – Wired networks are safer than wireless.
5. Distance – Longer cables may weaken the signal.
6. Maintenance – Some cables need more care.

## 8. ISO Seven-Layer Model (OSI Model)

This model helps understand how networks work in 7 layers:

1. Physical – Wires, cables, and signals.
2. Data Link – Ensures data is sent correctly.
3. Network – Finds the best path for data.
4. Transport – Ensures complete and error-free data.
5. Session – Keeps communication active.
6. Presentation – Converts data into a readable format.
7. Application – What users see (like websites, emails).

## 9. Twisted Pair Cable

Twisted pair cable has two copper wires twisted together to reduce interference and crosstalk.

### **UTP (Unshielded Twisted Pair):**

Used in homes and offices for internet connections.

Cheaper and easier to install.

Offers less protection from electrical noise.

### **STP (Shielded Twisted Pair):**

Used in industries and noisy environments.

Has extra shielding to block interference.

More expensive and slightly harder to install.

## 10. Coaxial Cable

Coaxial cable has one central copper wire, covered by insulation and shielding.

It protects signals from outside interference and supports high-frequency transmission.

Commonly used for TV connections, internet services, and CCTV camera setups.

## 11. Fiber Optic Cable

Fiber optic cables use light signals to transmit data, offering very high speed and long-distance communication.

### **Single-Mode Fiber (SMF):**

Used for long-distance communication like telephone and internet lines.

Has a very thin core allowing only one light signal to pass through.

Gives higher speed and less signal loss over long distances.

### **Multi-Mode Fiber (MMF):**

Used for short distances like inside buildings and LAN networks.

Has a thicker core allowing multiple light signals at once.

Cheaper than SMF but suffers more signal loss over distance.